AI FACTORY / INCIDENT REPORT

Synthetic bearing line | Educational decision support

Incident 6b2fdf9a-69bd-4c5a-b678-dee83e9448ea

Created 2026-09-06T10:05:55.473770+00:00

Supervisor status: MODIFIED

Sensor observation time: 2026-01-26 10:00:00+00:00

Machine M-01 | Failure probability: 87.2% in next 6 hours | Model: random_forest v1

Image classification: defect | Confidence: 99.9% | Domain: synthetic bearing surfaces

Prediction evidence

One-feature replacement with training median (local sensitivity; not causal)

temperature_mean_6: value 82.097, reference 70.371, probability change +0.340

vibration_mean_6: value 4.877, reference 3.351, probability change +0.290

temperature: value 85.390, reference 70.505, probability change +0.196

vibration: value 5.929, reference 3.360, probability change +0.167

pressure: value 5.537, reference 5.939, probability change +0.072

Maintenance note: Bearing inspection with vibration and quality review.

Retrieved evidence

bearing_sop.txt

SYNTHETIC TRAINING MANUAL — not an industrial procedure. Section 1: Vibration and bearing wear Persistent vibration accompanied by rising temperature may indicate bearing wear. Ask the supervisor to inspect the bearing and lubrication history. Compare multiple sensor samples before recommending maintenance. Only qualified personnel may perform maintenance.

bearing_sop.txt

Section 2: Surface quality Scratches and pits on bearing surfaces require supervisor review and segregation of the affected lot for inspection. Investigate whether quality loss coincides with vibration changes. The image classifier is trained on synthetic surfaces only.

bearing_sop.txt

Section 4: Reduced load A temporary reduction in machine load may lower stress while an inspection is arranged. The digital twin assumes 70 percent throughput and a hypothetical risk multiplier of 0.55; these assumptions require real-world validation.

Digital twin comparison

Action	Good units	Downtime h	Expected cost
continue	451.3	3.74	4280.62
reduce_load	440.9	2.06	3311.7
maintenance	576.5	2.56	2320.89

Illustrative assumptions: 8-hour horizon; 120 units/hour; failure repair cost 2,400; failure downtime 4 hours; lost-unit cost 4. Planned maintenance takes 2 hours and costs 450. Reduced load uses 70% throughput and 0.55 risk multiplier; maintenance uses 0.15 risk multiplier.

The 6-hour model probability converts to the scenario horizon using a hypothetical constant hazard. Reject rate: 11.7% (production record). Image confidence is not used as the batch reject rate. Monetary amounts are illustrative currency units.

Recommendation: maintenance

Lowest expected cost among three illustrative scenarios; human approval required. Image classifier flagged this item: supervisor quality review is required before release.

Human decision

Decision: modify

Final action: reduce_load

Reason: SYNTHETIC REHEARSAL, not a real supervisor decision

Audit ID: 649b8ea2-c331-4dd7-9a61-9576b3ab7c2c

Limitations

Models, images, sensor histories and manuals in this demo are synthetic. Probabilities are uncalibrated; the digital twin uses hypothetical costs and risk multipliers. Grad-CAM and feature sensitivity are explanations of model behavior, not proof of physical cause. No machinery commands are executed.